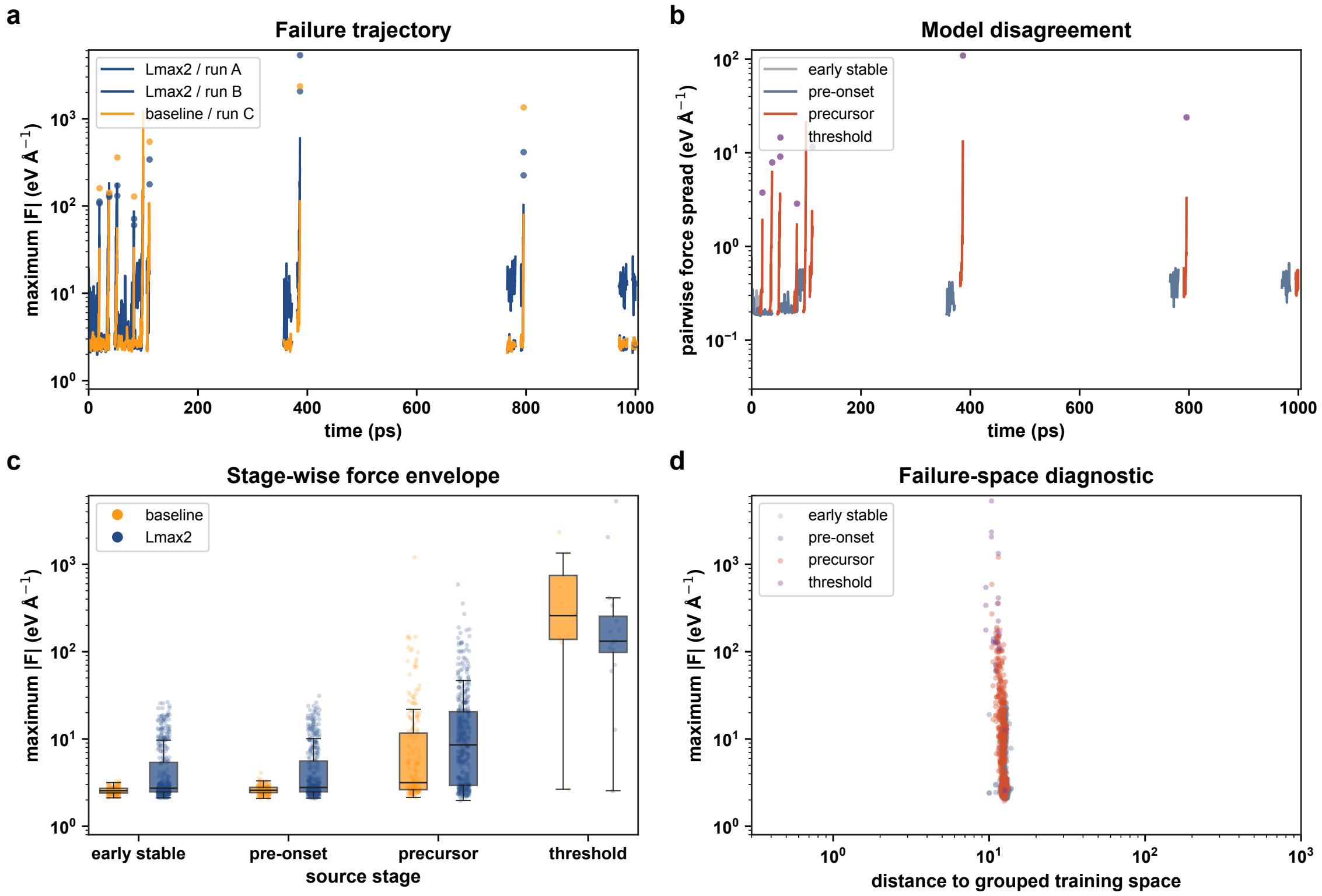

a
Failure trajectory
Lmax2 / run A
Lmax2 / run B
baseline / run C
maximum |F| (eV Å^{-1})
time (ps)
b
Model disagreement
early stable
pre-onset
precursor
threshold
pairwise force spread (eV Å^{-1})
time (ps)
c
Stage-wise force envelope
baseline
Lmax2
maximum |F| (eV Å^{-1})
early stable
pre-onset
precursor
threshold
source stage
d
Failure-space diagnostic
early stable
pre-onset
precursor
threshold
maximum |F| (eV Å^{-1})
distance to grouped training space